Integral diagonal methods for Sun's Conjecture 5.7(ii)

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Abstract

Let $W_n(1,t)=\sum_{k=0}^{\lfloor n/2\rfloor} t^k\binom{n}{2k}^2\binom{2k}{k}^2$. We give complete proofs of the two assertions in Conjecture 5.7(ii) of version 11 of Z.-W. Sun's 2010 preprint: for every positive integer N, the sum $\sum_{n<N}(42n+37)(-1)^nW_n(1,-16)$ is divisible by N, and for every prime p it is congruent to $p(21(\frac{p}{7})+16)$ modulo p^2. For the first assertion we extend an integral diagonal criterion to the denominator $1+u+v+\kappa uv$. At $\kappa=2$, its odd diagonals are controlled by Bing He's published divisibility theorem for a Ramanujan-type cubic-binomial sum. Exact reflected products and Abel summation establish every odd prime-power base. An integral two-state digit operator propagates these bases to arbitrary odd indices, while a termwise binomial divisibility handles every even index. A separate finite proof establishes the prime-square assertion, including $p=2,3,7$. Its connection with established results is made explicit. Historical priority is not asserted.

1 The unchanged problem and the scope of the proof

Write $b_k=\binom{2k}{k}$, and define

$$w_n=W_n(1,-16)=\sum_{k=0}^{\lfloor n/2\rfloor}(-16)^k\binom{n}{k}^2\binom{n-k}{k}^2=\sum_{k=0}^{\lfloor n/2\rfloor}(-16)^kb_k^2\binom{n}{2k}^2. \tag{1.1}$$

The equality follows from $\binom{n}{k}\binom{n-k}{k}=\binom{n}{2k}\binom{2k}{k}$. Set

$$S(N)=\sum_{n=0}^{N-1}(42n+37)(-1)^nw_n. \tag{1.2}$$

Theorem 1.1. *For every integer $N\geq 1$,*

$$N\mid S(N). \tag{1.3}$$

For every prime p,

$$S(p)\equiv p\left(21\left(\frac{p}{7}\right)+16\right)\pmod{p^2}, \tag{1.4}$$

where $(\frac{p}{7})$ is the Legendre symbol, including the value zero at $p=7$.

These are exactly the two assertions of Sun's Conjecture 5.7(ii) in [1, Conjecture 5.7(ii), p. 32]. We specify the version because later versions of that preprint reorganized the conjectures. The all-index statement includes every composite integer and every prime power; the second statement includes all primes, without an exceptional prime.

The principal arithmetic input for the all-index proof is He's published cubic-binomial divisibility [2], stated explicitly in Theorem 4.1. All formal-coordinate, valuation, interpolation,

and propagation steps needed to pass from that input to (1.3) are proved below. No complex multiplication, ordinary-prime hypothesis, numerical assumption, or conjectural supercongruence is used in that argument.

The prime-square assertion has a different status in the proof. We include a finite, self-contained proof of its arithmetic reduction and its cubic-binomial congruence. However, the cubic congruence was already known in stronger forms [3], and the weighted reduction can also be obtained from the published Apéry-polynomial results of Sun and Guo–Zeng [4, 5]. Thus the prime-square calculation is not presented as a historically new arithmetic principle.

2 A constant-term identity for the original sum

All formal series in this section have integral coefficients. Put

$$\Lambda = \frac{(1+u)(1+v)}{uv}, \qquad r = -\Lambda, \qquad \rho = \frac{uv}{1+u+v+2uv}, \qquad G(x) = \sum_{l\ge 0}(-16)^l b_l^2 x^{2l}.$$

Here CT denotes the coefficient of u^0v^0. Products of a Laurent polynomial with the displayed formal series have well-defined integral constant terms. Direct extraction gives

$$w_n = \mathrm{CT}\big(\Lambda^n G(uv)\big).$$

Since $(r-1)^{-1} = -\rho$, the finite geometric-sum identity and its formal derivative give

$$S(N) = \mathrm{CT}\big(r^N\Psi\big) - 42N\,\mathrm{CT}\big(r^N\rho G(uv)\big), \tag{2.1}$$

$$\Psi = (5\rho - 42\rho^2)G(uv). \tag{2.2}$$

For clarity, for arbitrary A, B one has

$$\sum_{n=0}^{N-1}(An+B)r^n = \frac{ANr^N}{r-1} + (r^N-1)\left(\frac{B}{r-1} - \frac{Ar}{(r-1)^2}\right).$$

Substitute $A = 42, B = 37$ and $r/(r-1)^2 = -\rho + \rho^2$. The terms without r^N have zero constant term after multiplication by $G(uv)$, proving (2.1).

Let $\theta_u = u\partial_u$ and $\theta_v = v\partial_v$. If

$$\Psi = \theta_u P + \theta_v Q, \qquad P, Q \in \mathbb{Z}[[u,v]], \tag{2.3}$$

then formal integration by parts shows that

$$\mathrm{CT}(r^N\Psi) = -N\,\mathrm{CT}\left(r^{N-1}\big((\theta_u r)P + (\theta_v r)Q\big)\right) \in N\mathbb{Z}.$$

Together with (2.1), this proves (1.3). The next two sections establish (2.3) with integral coefficients.

3 A diagonal criterion with a bilinear denominator

Put $A = \mathbb{Z}[[u,v]]$, $\theta_u = u\partial_u$, $\theta_v = v\partial_v$, and $\mathcal{D}_A = \theta_u A + \theta_v A$.

Theorem 3.1 (Integral diagonal criterion)**.** *Let $\kappa, a, b \in \mathbb{Z}$, let $R(x) \in \mathbb{Z}[[x]]$, and put*

$$\rho_\kappa = \frac{uv}{1+u+v+\kappa uv}, \qquad F(u,v) = (a\rho_\kappa + b\rho_\kappa^2)R(uv).$$

Then

$$F \in \mathcal{D}_A \quad\Longleftrightarrow\quad n \mid [u^n v^n]F \quad \text{for every } n \ge 1.$$

Proof. For any $F \in A$, coefficientwise Bezout gives

$$F \in \mathcal{D}_A \quad \Longleftrightarrow \quad [u^0v^0]F = 0 \quad \text{and} \quad \gcd(r,s) \mid [u^r v^s]F \quad ((r,s) \neq (0,0)).$$

The forward implication of the theorem follows immediately.

For the converse, let $C_\kappa(x) \in 1 + x\mathbb{Z}[[x]]$ be the unique solution of

$$C_\kappa = 1 + xC_\kappa^2 - \kappa x C_\kappa.$$

This recursion proves existence, uniqueness and integrality coefficient by coefficient. Define

$$U = uC_\kappa(uv), \qquad V = vC_\kappa(uv), \qquad y = UV.$$

This is an integral formal automorphism. More explicitly, let $B(y) \in 1 + y\mathbb{Z}[[y]]$ be the unique solution of

$$B^2 - (1+y)B + \kappa y = 0, \qquad \text{equivalently} \qquad B = 1 + y - \frac{\kappa y}{B}.$$

Again the latter equation proves integrality and uniqueness by recursion. The inverse coordinate change is

$$u = \frac{U}{B(y)}, \qquad v = \frac{V}{B(y)}, \qquad x = uv = \frac{y}{B(y)^2}, \qquad C_\kappa(x) = B(y).$$

Write $e(y) = yB'(y)/B(y)$. The inverse logarithmic differential matrix and its determinant are

$$\begin{pmatrix} d\log u \\ d\log v \end{pmatrix} = \begin{pmatrix} 1-e & -e \\ -e & 1-e \end{pmatrix} \begin{pmatrix} d\log U \\ d\log V \end{pmatrix}, \qquad J(y) = 1 - 2e(y) \in 1 + y\mathbb{Z}[[y]].$$

Both this matrix and its inverse have integral power-series entries. Integral exact logarithmic two-forms are therefore preserved in both directions. In other words, $F \in \mathcal{D}_A$ if and only if

$$\widetilde{F}(U,V) = J(y)F\left(\frac{U}{B(y)}, \frac{V}{B(y)}\right)$$

is an integral logarithmic divergence in U, V.

The defining equation for B gives the factorization

$$1 + u + v + \kappa uv = \frac{(1+U)(1+V)}{B(y)}, \qquad \rho_\kappa = \frac{y}{B(y)(1+U)(1+V)}.$$

Consequently

$$\widetilde{F} = \frac{A_1(y)}{(1+U)(1+V)} + \frac{A_2(y)}{(1+U)^2(1+V)^2},$$

where

$$A_1(y) = \frac{aJ(y)y}{B(y)} R\left(\frac{y}{B(y)^2}\right), \qquad A_2(y) = \frac{bJ(y)y^2}{B(y)^2} R\left(\frac{y}{B(y)^2}\right).$$

These are integral and start in degrees one and two respectively. Write $A_j(y) = \sum_k A_{j,k} y^k$. For $r, s \geq 1$, put $q = \min(r,s)$ and $g = \gcd(r,s)$. Then

$$[U^r V^s]\widetilde{F} = (-1)^{r+s} \sum_{k=0}^{q} (A_{1,k} + A_{2,k}(r-k+1)(s-k+1)).$$

Since r, s, q are multiples of g, this yields

$$[U^r V^s]\widetilde{F} \equiv (-1)^{r+s} [U^q V^q]\widetilde{F} \pmod{g}.$$

It remains to check the diagonal divisibility after substitution. Write $\operatorname{Diag} F(x) = \sum_n [u^n v^n] F\, x^n$. The radial substitution preserves the difference of the two exponents, so

$$\operatorname{Diag} \widetilde{F}(y) = J(y) \operatorname{Diag} F\left(\frac{y}{B(y)^2}\right).$$

By the hypothesis there is $H_0(x) \in x\mathbb{Z}[[x]]$ such that $\operatorname{Diag} F(x) = xH_0'(x)$. For $x(y) = y/B(y)^2$,

$$\frac{yx'(y)}{x(y)} = 1 - 2\frac{yB'(y)}{B(y)} = J(y),$$

and hence

$$\operatorname{Diag} \widetilde{F}(y) = y\frac{d}{dy} H_0(x(y)).$$

Its coefficient of y^q is divisible by q. The preceding coefficient congruence now gives all gcd conditions. Constant and axis coefficients vanish, so the Bezout criterion proves $\widetilde{F}$ is an integral logarithmic divergence. Pulling back proves the same for F. □

Lemma 3.2 (The diagonal generating function)**.** *With the preceding notation, put*

$$Q_\kappa(x) = (1 + \kappa x)^2 - 4x.$$

Then

$$\operatorname{Diag} F(x) = \frac{xR(x)\big(aQ_\kappa(x) + bx(1 + \kappa x)\big)}{Q_\kappa(x)^{3/2}}.$$

Proof. Expanding geometrically in $u + v$ gives

$$\operatorname{Diag} \frac{1}{1 + u + v + \kappa uv} = \sum_{i \geq 0} \binom{2i}{i} \frac{x^i}{(1 + \kappa x)^{2i+1}} = Q_\kappa(x)^{-1/2}.$$

The analogous expansion for the squared denominator gives $(1 + \kappa x)Q_\kappa(x)^{-3/2}$. Multiplication by uv, $(uv)^2$ and $R(uv)$ yields the formula. □

Corollary 3.3 (The signed kernel for Conjecture 5.7)**.** *Let*

$$G(x) = \sum_{\ell \geq 0} (-16)^\ell \binom{2\ell}{\ell}^2 x^{2\ell}, \quad \rho = \frac{uv}{1 + u + v + 2uv}, \quad \Psi = (5\rho - 42\rho^2)G(uv).$$

Then

$$\Psi \in \theta_u \mathbb{Z}[[u, v]] + \theta_v \mathbb{Z}[[u, v]] \quad \Longleftrightarrow \quad N \mid [x^{N-1}] \frac{G(x)(5 - 42x - 64x^2)}{(1 + 4x^2)^{3/2}} \quad (N \geq 1).$$

4 The integral diagonal

Write $b_j = \binom{2j}{j}$. The diagonal sequence needed for the signed constant-term reduction is

$$Y(2h + 1) = \sum_{i+l=h} (42i + 5)(-1)^i (-16)^l b_i b_l^2, \qquad h \geq 0, \tag{4.1}$$

$$Y(2M) = -42 \sum_{i+l=M-1} (2i + 1)(-1)^i (-16)^l b_i b_l^2, \qquad M \geq 1. \tag{4.2}$$

All indices in these sums are nonnegative integers. We prove the complete divisibility statement

$$N \mid Y(N) \qquad (N \geq 1).$$

4.1 An established Ramanujan partial-sum divisibility

The one arithmetic input is the following published theorem. It is not claimed as a new result here.

Theorem 4.1 (Bing He [2])**.** *For every integer $n > 1$,*

$$2n\binom{2n}{n} \;\Big|\; \sum_{k=0}^{n-1}(42k+5)\binom{2k}{k}^3 4096^{\,n-1-k}.$$

Define

$$A_j = \frac{(42j+5)b_j^3}{4096^j}, \qquad R_j = \sum_{k=0}^{j} A_k.$$

For every odd prime p and every $j \geq 0$,

$$v_p(R_j) \geq v_p\big((2j+1)(2j+2)\big). \tag{4.3}$$

Indeed, for $j \geq 1$, apply Theorem 4.1 with $n = j+1$ and use the exact equality

$$2(j+1)b_{j+1} = 4(2j+1)b_j.$$

The denominator 4096^j is a p-adic unit, and the Catalan integrality $b_j/(j+1) \in \mathbb{Z}$ gives the bound. For $j = 0$, it follows from $R_0 = 5$ and the fact that p is odd.

4.2 Every odd prime-power base

Theorem 4.2. *For every odd prime p and every integer $e \geq 1$,*

$$p^e \mid Y(p^e).$$

Proof. Put $N = p^e$, $h = (N-1)/2$, and

$$r_j = \prod_{a=1}^{j}\left(\frac{1 - N/(2a-1)}{1 - N/(2a)}\right)^2 \quad (0 \leq j \leq h), \qquad r_0 = 1.$$

The factors are p-adic units: all positive denominators $2a-1, 2a$ are less than N, so their valuations are smaller than e. Comparing consecutive ratios, starting at $j = 0$, gives the exact identity

$$\left(\frac{b_{h-j}}{b_h}\right)^2 = \frac{b_j^2}{256^j}\, r_j.$$

Substitution $l = h - j$, $i = j$ into (4.1) therefore gives

$$Y(N) = (-16)^h b_h^2 \sum_{j=0}^{h} A_j r_j. \tag{4.4}$$

For $0 \leq j < h$, set $a = 2j+1$, $b = 2j+2$. Direct subtraction of the consecutive products gives

$$r_{j+1} - r_j = -N r_j \frac{2ab - N(a+b)}{a^2(b-N)^2}.$$

The integers a, b are consecutive and smaller than N. If $t = v_p(ab)$, then $t < e$, $v_p(b-N) = v_p(b)$, and $v_p(2ab - N(a+b)) = t$. The last equality uses that p is odd: the first term has valuation t, and the second has valuation at least e. It follows that

$$v_p(r_{j+1} - r_j) = e - v_p\big((2j+1)(2j+2)\big).$$

Abel summation now yields

$$\sum_{j=0}^{h} A_j r_j = R_h r_h - \sum_{j=0}^{h-1} R_j (r_{j+1} - r_j) \in p^e \mathbb{Z}_p.$$

Here each term of the last sum has valuation at least e by (4.3), while $R_h \in p^e \mathbb{Z}_p$ because $2h + 1 = N$. The integer prefactor in (4.4) proves the theorem. This argument includes $p = 3$ and $p = 7$. □

4.3 Integral central-binomial digit kernels

We give the integral interpolation needed to propagate these bases; the proof does not discard carries. Fix an arbitrary odd prime p. The ring $\mathbb{Z}_p\langle X_1, \ldots, X_d\rangle$ consists of power series whose coefficients lie in $\mathbb{Z}_p$ and tend to zero as total degree tends to infinity. It is complete for the Gauss norm. Integral affine substitutions preserve the ring, and evaluation at points of $\mathbb{Z}_p^d$ is well defined.

Lemma 4.3 (Normalized unit factorials)**.** *Put*

$$E_p(M) = \prod_{\substack{1 \le j \le M \\ p \nmid j}} j, \qquad W = (p-1)!.$$

For each $0 \le r < p$ there is a unit $\mathcal{H}_r(X) \in 1 + p\mathbb{Z}_p\langle X\rangle$, with $\mathcal{H}_r(0) = 1$, such that

$$E_p(pq + r) = W^q r!\, \mathcal{H}_r(q) \qquad (q \ge 0).$$

Proof. Set

$$B(X) = \prod_{i=1}^{p-1} (1 + pX/i), \qquad \log B(X) = \sum_{j \ge 1} \ell_j X^j, \quad \ell_j = \frac{(-1)^{j-1} p^j}{j} \sum_{i=1}^{p-1} i^{-j}.$$

The polynomial continuation of $\sum_{q=0}^{n-1} q^j$ is

$$S_j(X) = \sum_{a=0}^{j} \left\{ {j \atop a} \right\} \frac{(X)_{\underline{a+1}}}{a+1},$$

where the braces denote Stirling numbers of the second kind and the underlined subscript denotes a falling factorial. In particular, $S_j \in p^{-\lfloor \log_p(j+1) \rfloor} \mathbb{Z}_p[X]$. For every odd p and $j \ge 1$,

$$j - v_p(j) - \lfloor \log_p(j+1) \rfloor \ge 1,$$

and these bounds tend to infinity with j. The inequality is immediate for $j = 1, 2, 3$. For $j \ge 4$, use $v_p(j) \le \lfloor \log_3(j+1) \rfloor$ and $2\lfloor \log_3(j+1) \rfloor \le j - 1$; the latter follows directly for $4 \le j \le 7$, and then from $2t \le 3^t - 2$ when $t = \lfloor \log_3(j+1) \rfloor \ge 2$. Consequently,

$$A(X) = \sum_{j \ge 1} \ell_j S_j(X) \in p\mathbb{Z}_p\langle X\rangle.$$

Since p is odd, $\exp A(X)$ converges in $1 + p\mathbb{Z}_p\langle X\rangle$: its k-th term has valuation at least $k - v_p(k!)$, which tends to infinity. Define

$$\mathcal{H}_r(X) = \exp A(X) \prod_{i=1}^{r} (1 + pX/i).$$

All $S_j(0)$ vanish, giving $\mathcal{H}_r(0) = 1$. For every nonnegative integer q, convergence gives

$$A(q) = \sum_{a=0}^{q-1} \log B(a), \qquad \exp A(q) = \prod_{a=0}^{q-1} B(a).$$

Splitting $E_p(pq + r)$ into its complete blocks and its last r factors proves the assertion. Only the normalized factor is interpolated; no interpolation of W^q is needed. □

Lemma 4.4 (Central-binomial kernel)**.** *For $0 \le r < p$, put*

$$\delta_r = \left\lfloor \frac{2r}{p} \right\rfloor, \qquad t_r = 2r - p\delta_r,$$

and define

$$\mathcal{B}_r(X) = p^{\delta_r}(2X+1)^{\delta_r} W^{\delta_r} \frac{t_r!}{r!^2} \frac{\mathcal{H}_{t_r}(2X + \delta_r)}{\mathcal{H}_r(X)^2}.$$

Then $\mathcal{B}_r \in p^{\delta_r}\mathbb{Z}_p\langle X\rangle$, and for all $q \ge 0$,

$$b_{pq+r} = b_q \mathcal{B}_r(q), \qquad \mathcal{B}_r(0) = b_r, \qquad \mathcal{B}_r(X) \equiv b_r \pmod{p}.$$

Proof. All factorial denominators displayed in $\mathcal{B}_r$ are units. Use the identity

$$M! = p^{\lfloor M/p \rfloor} \lfloor M/p \rfloor! E_p(M)$$

in b_{pq+r}, followed by Lemma 4.3. The factorial part remaining after removing the unit factorials is

$$p^{\delta_r} \frac{(2q + \delta_r)!}{q!^2} = p^{\delta_r} b_q (2q+1)^{\delta_r},$$

and the powers of W reduce to W^{δ_r}. This proves the exact formula. Evaluation at $q = 0$ gives the constant value. If $\delta_r = 1$, both sides of the asserted congruence vanish modulo p; if $\delta_r = 0$, all $\mathcal{H}$'s are one modulo p, and $t_r = 2r$, proving the congruence. □

Put

$$w(i,l) = (-1)^i (-16)^l b_i b_l^2, \qquad \mathcal{A}_p = \mathbb{Z}_p\langle X, Y\rangle.$$

The restricted series

$$\mathcal{C}(Y) = \exp\bigl(Y \log((-16)^{p-1})\bigr) \in 1 + p\mathbb{Z}_p\langle Y\rangle$$

satisfies $\mathcal{C}(j) = (-16)^{(p-1)j}$ for integers $j \ge 0$. Define

$$K_{r,s}(X,Y) = (-1)^r (-16)^s \mathcal{C}(Y) \mathcal{B}_r(X) \mathcal{B}_s(Y)^2. \tag{4.5}$$

Because p is odd, $(-1)^{pq+r} = (-1)^{q+r}$, so Lemma 4.4 gives the exact factorization

$$w(pq + r, pj + s) = w(q,j) K_{r,s}(q,j) \qquad (q, j \ge 0). \tag{4.6}$$

It also gives

$$\begin{gathered} K_{r,s} \in p^{\delta_r + 2\delta_s} \mathcal{A}_p, \qquad K_{r,s}(0,0) = w(r,s), \\ K_{r,s}(X,Y) \equiv w(r,s) \pmod{p}. \end{gathered} \tag{4.7}$$

4.4 A two-state operator on odd indices

For $F = (F_0, F_1) \in \mathcal{M}_p := \mathcal{A}_p^2$ and every positive odd integer N, define

$$\mathcal{J}_N(F) = \sum_{i+l=(N-1)/2} w(i,l)F_0(i,l) + \sum_{i+l=(N-1)/2-1} w(i,l)F_1(i,l).$$

A sum with a negative right-hand side is empty; in particular, $\mathcal{J}_1(F) = F_0(0,0)$. Let $h_p = (p-1)/2$.

Proposition 4.5 (Exact digit decomposition)**.** *For $d = 0, 1$, set*

$$(\mathcal{L}_p F)_d(X,Y) = \sum_{a=0}^{1} \sum_{\substack{0 \le r,s < p \\ r+s = h_p - a + pd}} K_{r,s}(X,Y) F_a(pX + r, pY + s).$$

Then $\mathcal{L}_p$ maps $\mathcal{M}_p$ to itself, and for every positive odd integer N,

$$\mathcal{J}_{pN}(F) = \mathcal{J}_N(\mathcal{L}_p F).$$

Moreover, there is $R_F \in \mathcal{M}_p$ with

$$\mathcal{L}_p F = (\mathcal{J}_p(F), 0) + pR_F, \qquad (R_F)_0(0,0) = 0.$$

Proof. In the input state $a \in \{0,1\}$, write $i = pq + r$, $l = pj + s$, $0 \le r, s < p$. The constraint becomes

$$p(q+j) + r + s = \frac{pN-1}{2} - a = p\frac{N-1}{2} + h_p - a.$$

Thus

$$r + s = h_p - a + pd, \qquad q + j = \frac{N-1}{2} - d.$$

Only $d = 0, 1$ is possible: $h_p - a \ge 0$, and $0 \le r + s \le 2p - 2$. This includes $p = 3$, when the smallest value $h_p - a$ is zero. Equation (4.6) proves the exact block identity, with all empty ranges respected. Integral affine substitution and the finite sum preserve $\mathcal{M}_p$.

In output state $d = 1$,

$$r + s = p + h_p - a \ge p + h_p - 1 \ge p > p - 1.$$

If neither central binomial coefficient carried, then $r, s \le h_p$, contradicting this inequality. Hence $\delta_r + 2\delta_s \ge 1$, and $(\mathcal{L}_p F)_1 \in p\mathcal{A}_p$.

For $d = 0$, use (4.7) and

$$F_a(pX + r, pY + s) \equiv F_a(r, s) \pmod{p}.$$

The latter holds first for polynomials and then for restricted power series by Gauss-norm continuity. The resulting constant is exactly $\mathcal{J}_p(F)$. Moreover, evaluating the first component at $(X, Y) = (0, 0)$ gives this constant exactly, because $K_{r,s}(0,0) = w(r,s)$. Subtracting the constant vector and dividing by p proves the last assertion, including its zero constant term. □

Theorem 4.6 (Propagation from prime-power bases)**.** *Let p be an odd prime, $F \in \mathcal{M}_p$, and $e \ge 1$. Suppose that*

$$\mathcal{J}_{p^j}(F) \in p^j \mathbb{Z}_p \qquad (1 \le j < e).$$

Then, for every positive odd integer m,

$$\mathcal{J}_{p^e m}(F) \equiv \mathcal{J}_{p^e}(F)\mathcal{J}_m(1,0) \pmod{p^e}.$$

If the base divisibility also holds at $j = e$, then $\mathcal{J}_{p^e m}(F) \in p^e \mathbb{Z}_p$ for every such m. No coprimality condition on m is required.

Proof. Iterating the exact block identity gives

$$(\mathcal{L}_p^j F)_0(0,0) = \mathcal{J}_{p^j}(F).$$

Suppose the first $j-1$ base divisibilities have given $\mathcal{L}_p^{j-1}F = p^{j-1}G$, $G \in \mathcal{M}_p$; for $j = 1$, take $G = F$. Proposition 4.5 yields

$$\mathcal{L}_p^j F = p^{j-1}(\mathcal{J}_p(G), 0) + p^j R_G = (\mathcal{J}_{p^j}(F), 0) + p^j R_G.$$

The second equality follows either from the exact block identity or by evaluation at $(0,0)$, using $(R_G)_0(0,0) = 0$. The next base divisibility therefore puts the whole iterate in $p^j\mathcal{M}_p$. Induction through $e-1$, and then one more application of the formula, gives

$$\mathcal{L}_p^e F = (\mathcal{J}_{p^e}(F), 0) + p^e R \qquad (R \in \mathcal{M}_p).$$

Apply $\mathcal{J}_m$, which is integral on $\mathcal{M}_p$. The hypothesis is empty for $e = 1$, and the same argument applies. The case $m = 1$ is included throughout. □

Corollary 4.7. *For every positive odd integer N, one has $N \mid Y(N)$.*

Proof. Take $F = (42X + 5, 0)$, so $\mathcal{J}_N(F) = Y(N)$ for odd N. Theorem 4.2 supplies every base required by Theorem 4.6. For any prime power $p^e \mid N$, the cofactor $m = N/p^e$ is odd; hence $p^e \mid Y(N)$. This proves the assertion for all odd $N > 1$. For $N = 1$, it is immediate from $Y(1) = 5$. Exponent zero requires no divisibility condition. □

4.5 Even indices: an integral individual summand

Lemma 4.8. *For all nonnegative integers i, l,*

$$i + l + 1 \;\Big|\; (2i+1)b_i b_l.$$

Proof. Set $M = i + l + 1$, and fix $p^e \mid M$ with $e \geq 1$. For $1 \leq j \leq e$, put $q = p^j$, and let r, s be the residues of i, l modulo q, respectively. Then $r + s = q - 1$: their sum is congruent to -1 modulo q, and lies between zero and $2q - 2$. The contribution of this level to $v_p(b_i) + v_p(b_l)$ is

$$\left\lfloor \frac{2r}{q} \right\rfloor + \left\lfloor \frac{2s}{q} \right\rfloor.$$

If q is even, this equals one. If q is odd, it also equals one unless $r = s = (q-1)/2$; in that exceptional case it is zero, but $q \mid 2i + 1$. Legendre's factorial valuation formula and $v_p(2i+1) = \sum_{j \geq 1} \mathbf{1}_{p^j \mid 2i+1}$ show that the total valuation of $(2i+1)b_i b_l$ is at least e. This holds for every prime power dividing M, proving the lemma. □

Theorem 4.9 (Complete diagonal integrality)**.** *For every integer $N \geq 1$,*

$$N \mid Y(N).$$

Proof. Odd indices are covered by Corollary 4.7. For $N = 2M$, each summand in (4.2) is divisible by $2M$: Lemma 4.8 supplies the factor M, the remaining factor b_l and all powers are integers, and the coefficient 42 supplies the factor two. This includes $M = 1$. □

5 Completion of the all-index assertion

The odd and even formulas (4.1) and (4.2) are the coefficients of the diagonal generating function in Corollary 3.3. Indeed,

$$(1+4z)^{-3/2} = \sum_{i\geq 0} (-1)^i (2i+1) b_i z^i,$$

and

$$[z^i] \frac{5-64z}{(1+4z)^{3/2}} = (-1)^i (42i+5) b_i.$$

For $i \geq 1$, the latter identity uses $b_{i-1}/b_i = i/(2(2i-1))$; for $i = 0$, both sides are five. The odd term $-42x$ in the numerator gives exactly (4.2).

Theorem 4.9 proves every diagonal divisibility required by Corollary 3.3. The integral diagonal criterion therefore supplies P, Q in (2.3). Applying the integration-by-parts identity of Section 2 proves (1.3) for every $N \geq 1$. This completes the first assertion of Theorem 1.1.

6 Sun's signed weighted congruence modulo a prime square

Put $b_k = \binom{2k}{k}$ and

$$W_n(1,t) = \sum_{k=0}^{\lfloor n/2 \rfloor} t^k b_k^2 \binom{n}{2k}^2.$$

We prove the unchanged prime-square assertion of Conjecture 5.7(ii): for every prime p,

$$\sum_{n=0}^{p-1} (42n+37)(-1)^n W_n(1,-16) \equiv p\left(21\left(\frac{p}{7}\right)+16\right) \pmod{p^2}. \tag{6.1}$$

Here $(p/7)$ is the Legendre symbol, with value zero when $p = 7$.

Sun's unweighted transformation and Guo–Zeng's weighted Apéry-polynomial congruence [4, 5] also give the signed transfer by an elementary index shift and coefficientwise differentiation. We give a direct alternative derivation through an alternating polynomial functional. The cubic auxiliary congruence used here follows from the stronger result of Sun [3, Theorem 1.3, (1.15)]; we include a finite algebraic proof to keep the argument independent of any supercongruence input. None of these observations establishes historical priority for the original W-congruence.

6.1 An alternating polynomial functional

For $f \in \mathbb{Q}[x]$, let $Q_f \in \mathbb{Q}[x]$ be the unique polynomial satisfying

$$Q_f(x+1) + Q_f(x) = 2f(x).$$

The operator $Q \mapsto Q(x+1) + Q(x)$ acts triangularly on monomials, with diagonal entries 2, proving existence and uniqueness. If p is odd and $f \in \mathbb{Z}_{(p)}[x]$, the same triangular argument shows $Q_f \in \mathbb{Z}_{(p)}[x]$. For every odd positive integer N, telescoping gives the exact identity

$$\sum_{n=0}^{N-1} (-1)^n f(n) = \frac{Q_f(0) + Q_f(N)}{2}. \tag{6.2}$$

In particular, for a p-integral polynomial f and an odd prime p, the right side is $Q_f(0) + (p/2)Q_f'(0)$ modulo p^2.

Lemma 6.1. *Let $f_m(x) = \binom{x}{m}^2$ and $Q_m = Q_{f_m}$. For every $k \geq 0$,*

$$Q_{2k}(0) = \frac{(-1)^k b_k}{16^k}, \qquad Q'_{2k}(0) = 0, \qquad Q'_{2k+1}(0) = \frac{(-1)^{k+1}}{(2k+1)b_k}.$$

Proof. For a polynomial f, the series $F_f(z) = \sum_{n\geq 0} f(n)z^n$ is a rational function whose only possible finite pole is at $z = 1$. Summing the defining relation for Q_f gives

$$(1+z)F_{Q_f}(z) - Q_f(0) = 2zF_f(z).$$

Evaluation at $z = -1$ therefore yields $Q_f(0) = 2F_f(-1)$. The elementary binomial generating identity

$$\sum_{n\geq 0} \binom{n}{m}^2 z^n = \frac{z^m}{(1-z)^{2m+1}} \sum_{j=0}^{m} \binom{m}{j}^2 z^j$$

follows from binomial product linearization and the binomial theorem. At $m = 2k$, use

$$\sum_{j=0}^{2k} (-1)^j \binom{2k}{j}^2 = (-1)^k \binom{2k}{k},$$

obtained by taking the coefficient of X^{2k} in $(1-X)^{2k}(1+X)^{2k}$. The first asserted formula follows.

The identity $f_m(m-1-x) = f_m(x)$ and uniqueness imply $Q_m(m-x) = Q_m(x)$. The double zeros of f_m at $0, 1, \ldots, m-1$ give, for $0 \leq j \leq m$,

$$Q_m(j) = (-1)^j Q_m(0), \qquad Q'_m(j) = (-1)^j Q'_m(0).$$

For even m, differentiation of the reflection identity gives $Q'_m(m) = -Q'_m(0)$, while the recurrence gives equality. Thus $Q'_m(0) = 0$. The case $m = 0$ is immediate.

For odd $m = 2k+1$, reflection of the values gives $Q_m(0) = 0$, and consequently Q_m vanishes at all $j = 0, \ldots, m$. Put $P(x) = \prod_{j=0}^{m}(x-j)$. The proper rational function $Q_m(x)/P(x)^2$ has only simple poles, with residues

$$\frac{Q'_m(j)}{P'(j)^2} = \frac{(-1)^j Q'_m(0)}{j!^2 (m-j)!^2}.$$

Since Q_m has degree $2m$ and leading coefficient $1/m!^2$, the coefficient of x^{-2} at infinity is $1/m!^2$. Comparing with the partial-fraction expansion yields

$$1 = Q'_m(0) \sum_{j=0}^{m} (-1)^j j \binom{m}{j}^2.$$

For $m = 2k+1$, the sum on the right is $(-1)^{k+1}(2k+1)b_k$. To see this, use $j\binom{m}{j} = m\binom{m-1}{j-1}$ and take the coefficient of X^{m-1} in $(1-X)^{m-1}(1+X)^m = (1-X^2)^{2k}(1+X)$. This proves the remaining derivative formula. □

6.2 The two signed inner sums

Lemma 6.2. *Let p be odd and $0 \leq 2k < p$. Then, in $\mathbb{Z}_{(p)}$,*

$$\sum_{n=0}^{p-1} (-1)^n \binom{n}{2k}^2 \equiv \frac{(-1)^k b_k}{16^k} \pmod{p^2},$$

$$\sum_{n=0}^{p-1} n(-1)^n \binom{n}{2k}^2 \equiv \left(k - \frac{1}{2}\right) \frac{(-1)^k b_k}{16^k} + \frac{p}{2} \frac{(-1)^k}{b_k} \pmod{p^2}.$$

Proof. The polynomial f_{2k} has p-integral coefficients because $(2k)!$ is a unit. Its value in (6.2), together with Lemma 6.1, proves the first congruence.

For the second, put $M_m = Q_{xf_m}$. The polynomial identity

$$(2x - m + 1)f_m(x) = (m + 1)\big(f_{m+1}(x + 1) - f_{m+1}(x)\big)$$

and the defining equation for Q give

$$2M_m - (m - 1)Q_m = 2(m + 1)(f_{m+1} - Q_{m+1}).$$

At $m = 2k$, one has $Q_{m+1}(0) = 0$ and $f_{m+1}(0) = f'_{m+1}(0) = 0$. Hence

$$M_{2k}(0) = \left(k - \frac{1}{2}\right) Q_{2k}(0), \qquad M'_{2k}(0) = \frac{(-1)^k}{b_k}.$$

Apply (6.2) to xf_{2k}.

This proof includes the endpoint $2k = p - 1$. Although the rational derivation passes through $f_{2k+1} = f_p$ there, coefficientwise p-integrality is needed only for f_{2k} and xf_{2k}, whose denominators are units. Also b_k is a unit whenever $2k < p$. □

Theorem 6.3 (General signed transfer)**.** *Let p be an odd prime, $h = (p-1)/2$, and $A, B, t \in \mathbb{Z}$. Then*

$$\sum_{n=0}^{p-1}(An + B)(-1)^n W_n(1, t) \equiv \sum_{k=0}^{h}\left(Ak + B - \frac{A}{2}\right)\left(-\frac{t}{16}\right)^k b_k^3 + \frac{A}{2}p\left(\frac{1 + 4t}{p}\right) \pmod{p^2}. \tag{6.3}$$

Proof. Interchange the finite sums and apply Lemma 6.2. The remaining term is

$$\frac{A}{2}p\sum_{k=0}^{h}(-t)^k b_k.$$

Since $b_k \equiv (-4)^k\binom{h}{k} \pmod{p}$, the sum is $(1 + 4t)^h$ modulo p. Euler's criterion, including the case $p \mid 1 + 4t$, gives the stated character. □

At $A = 42$, $B = 37$, $t = -16$, the first sum in (6.3) is

$$K_h = \sum_{k=0}^{h}(42k + 16)b_k^3.$$

For every odd $p \neq 3$, quadratic reciprocity gives $(-63/p) = (-7/p) = (p/7)$, with both sides zero at $p = 7$. At $p = 3$, the replacement of $(-63/p)$ by $(p/7)$ is still valid in (6.3), since its multiplier $21p$ is divisible by p^2. Therefore (6.1) is reduced, for every odd prime, to $K_h \equiv 16p \pmod{p^2}$.

6.3 A finite triangular WZ certificate

For $n \geq 0$ and $0 \leq k \leq n$, put

$$H(n, k) = (-1)^k \frac{(2n)!^2(2n - k)!}{n!^3(n - k)!^2(n + k)!} = b_n^3 \frac{(-n)_k^2}{(n + 1)_k(-2n)_k}.$$

At $(n,k)=(0,0)$ use the factorial expression, and set $H=0$ outside the triangular range. Define

$$D(n,k)=\frac{4(21n^2-14nk+k^2+8n-4k)}{2n-k}H(n,k),\qquad E(n,k)=\frac{16(2n+1)^2}{n+k+1}H(n,k),$$

with $D(0,0)=16$, and with D,E zero outside the triangle. The certificate is

$$D(n,k+1)-D(n,k)=E(n-1,k)-E(n,k). \tag{6.4}$$

For $0\le k<n$, it follows by substituting the ratios

$$\frac{H(n,k+1)}{H(n,k)}=-\frac{(n-k)^2}{(n+k+1)(2n-k)},$$

$$\frac{H(n-1,k)}{H(n,k)}=\frac{n(n-k)^2(n+k)}{4(2n-1)^2(2n-k)(2n-k-1)}$$

and clearing denominators in the resulting rational identity. At $k=n\ge 1$, the certificate reduces to

$$D(n,n)=E(n,n)=16(2n+1)H(n,n).$$

At $(0,0)$ it is immediate from the definitions. Beyond the triangle both sides vanish. Thus every boundary is included.

Summing (6.4) over $0\le n,k\le h$, using $D(n,h+1)=0$ and $E(-1,k)=0$, gives the exact finite identity

$$\begin{aligned}\sum_{n=0}^{h}(42n+16)b_n^3&=\sum_{k=0}^{h}E(h,k)\\&=16(2h+1)^2b_h^3\sum_{k=0}^{h}\frac{(-h)_k^2}{(h+1)_k(-2h)_k(h+k+1)}.\end{aligned} \tag{6.5}$$

The pair is motivated by reflection of Guillera's second WZ pair for Identity 2 in [6], associated with the $(42n+5)/64^n$ Ramanujan formula. The displayed certificate has been proved algebraically, so no analytic continuation is used.

Now take $h=(p-1)/2$. For $k<h$, every displayed denominator in (6.5) is a p-adic unit, so the corresponding $E(h,k)$ lies in $p^2\mathbb{Z}_{(p)}$. At $k=h$,

$$\frac{(-h)_h^2}{(h+1)_h(-2h)_h}=\frac{(-1)^h}{b_h^2},\qquad E(h,h)=16p(-1)^hb_h\equiv 16p\pmod{p^2},$$

because $b_h=\binom{p-1}{h}\equiv(-1)^h\pmod p$. Thus $K_h\equiv 16p\pmod{p^2}$ for every odd prime.

6.4 Completion and the prime two

Combining the signed transfer with the last congruence proves (6.1) for all odd primes, including 3 and 7. For $p=2$, the original two-term sum is $37-79=-42\equiv 2\pmod 4$, while the proposed right side is $2(21(2/7)+16)=74\equiv 2\pmod 4$. Hence every prime is covered.

This argument concerns only the prime-square assertion. It does not, on its own, prove the all-integer divisibility assertion or the unweighted assertion in Conjecture 5.7(i).

7 Scope, prior results, and verification

The proof establishes both assertions of the unchanged Conjecture 5.7(ii). The new all-index argument developed here has two separate arithmetic tasks: proving the bases $p^e \mid Y(p^e)$, and proving that those bases imply $p^e \mid Y(p^e m)$ for every odd cofactor m. The exact digit kernel and its two neighboring coefficient states are essential for the latter task. Proving only the first task would not establish the original all-index conjecture.

The formal coordinate change by itself does not prove the result. It converts a two-variable integrality problem into a precise divisibility condition. He's partial-sum theorem pays exactly for the denominators in the reflected-product differences; the digit operator then preserves that divisibility through every carry. The even diagonals are dealt with at all prime powers, including the prime two, by the separate termwise lemma.

For the prime-square assertion, the finite triangular identity supplies an independent proof of the required known cubic congruence. Its earlier occurrence in [3] is explicitly acknowledged. Sun's polynomial transformation [4, Theorem 1.1(i)] and Guo–Zeng's weighted Apéry-polynomial congruence [5, Theorem 1.1(ii)] also yield the weighted transfer after an index shift and coefficientwise differentiation. Consequently this part should not be used as evidence for a previously unknown general method.

A targeted literature check of the original $W_n(1,-16)$ all-index assertion, its weights, and OEIS A179537 did not locate an earlier proof of precisely (1.3). Such a search cannot certify absence from the literature. This manuscript makes no claim of historical firstness, and does not claim that every novelty or priority condition has been settled. It also does not claim a solution of the remaining ordinary-prime branches of Sun's separate Conjecture 5.8(i).

The proof is symbolic. Exact computations were used only to check transcription and boundary formulas: the original and diagonal divisibilities for $1 \le N \le 300$; the termwise even-index lemma for $0 \le i, l \le 60$; the exact reflection for odd $N \le 39$; the two-state identities for $p = 3, 5, 7, 11$ and odd cofactors $1, 3, 5, 7$, using three polynomial state pairs; and independent bivariate diagonal extraction through $N = 44$. The prime-square check covered every prime through 131. These finite checks are not a substitute for the unrestricted proofs above.

A A small reproducible check of the two original statements

The following Python code uses exact integers and no third-party packages. It checks a bounded range only. The all-quantifier arguments are contained in the body of the paper.

```
from math import comb, isqrt

def is_prime(p):
    return p >= 2 and all(p % d for d in range(2, isqrt(p) + 1))

def chi7(p):
    r = p % 7
    return 0 if r == 0 else (1 if r in (1, 2, 4) else -1)

def W(n):
    return sum((-16)**k * comb(n, 2*k)**2 * comb(2*k, k)**2
               for k in range(n//2 + 1))

total = 0
for N in range(1, 301):
    n = N - 1
    total += (42*n + 37) * (-1)**n * W(n)
    assert total % N == 0
    if is_prime(N):
```

```
        assert (total - N*(21*chi7(N) + 16)) % (N*N) == 0
print("PASS: original divisibility and prime-square checks")
```

References

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[3] Z.-W. Sun, Super congruences and Euler numbers, *Sci. China Math.* **54** (2011), 2509–2535, Theorem 1.3, equation (1.15). https://arxiv.org/abs/1001.4453.

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